

Event Information:

EventID :
39
Event Time :
Jan, 01, 2021, 04:45 PM
Rule :
SOC109 - Emotet Malware Detected
Level :
Security Analyst
Source Address :
172.16.17.83
Source Hostname :
Maxim
File Name :
MES 2020_12_31 S632974.doc
File Hash :
eee99e6d8ade9463dd206dfbad3485ea
File Size :
161.36 Kb
Device Action :
Allowed

Analysis Steps:

Logs were searched with IP 172[.]16.17.83 (Maxim) as the source in the relevant time-frame yielding 3 events. Unusual GET requests can be seen being made to three unique external IPs for each event. It is unclear from available logs how the file was first delivered.

External IP	Requested URL
23[.]111.174.153	hxxp://decpak[.]com/cgi-bin/gU/
152[.]170.79.100	hxxp://152[.]170.79.100/076ay2uof/umojx2x1vf1qjjk/hue4e670x/d3eobn8z0k0rp/syaxabx0loj/erz7hayf/
190[.]247.139.101	hxxp://190[.]247.139.101/i3u3l3e3n96/sow63klj/pkv3runqw/dudwqjl4zg8l7hk6ah6/huulsuajy/nibyefksabf1mz63/

On VirusTotal the the file is listed as malicious by 51/63 Vendors and also supports the alert's assertion that this is of the Emotet family of malware.

The screenshot shows the VirusTotal analysis interface for the file 'MES 2020_12_31 S632974.doc'. At the top, a red circle indicates that 51 out of 63 security vendors have flagged the file as malicious. Below this, the file's SHA-256 hash is displayed: 7045070957b656283751c3a7b72b6b023d6d86f848201ee85947392e27c. The file size is 161.36 KB, and it was last analyzed 3 months ago. The interface includes tabs for Detection, Details, Relations, Behavior, and Community. Under the 'Code insights' section, it notes that the provided macros do not exhibit any signs of malicious intent. The 'Crowdsourced AI' section shows a warning from Hopsaic flags this file as malicious. The 'Popular threat label' is 'downloader.emotet/w97m'. The 'Threat categories' are 'downloader' and 'trojan'. The 'Family labels' are 'emotet', 'w97m', and 'emotet'. The 'Security vendors' analysis' section shows a table of vendor detections:

Vendor	Detection
Acronis (Static ML)	Suspicious
AlhrtLab-V3	Downloader/DOC.Emotet.S1328
AliCloud	Trojan:MSOffice/Downloader.afu
ALTac	Trojan.Downloader.DOC.Gen

Joe's Sandbox also lists the file as malicious with 100% confidence.

Overview

General Information

Sample Name:

MES 2020_12_31
S632974.doc.vir (renamed file
extension from vir to doc)

Analysis ID:

648462

MD5:

eee99e6d8ade946...

SHA1:

204f560829a6d70f...

SHA256:

70450709057b656...

Tags:

doc docx

Infos:

Detection

MALICIOUS

SUSPICIOUS

CLEAN

UNKNOWN

Score:

100

Range:

0 - 100

Whitelisted:

false

Confidence:

100%

Signatures

Multi AV Scanner detection for submitted file

Office document tries to convince victim t...

Antivirus / Scanner detection for submitte...

Antivirus detection for URL or domain

Multi AV Scanner detection for domain / U...

PowerShell case anomaly found

Encrypted powershell cmdline option found

Very long command line found

Creates processes via WMI

Suspicious powershell command line found

Machine Learning detection for sample

Document contains an embedded VBA wi...

Document contains an embedded VBA m...

Obfuscated command line found

Potential dropper URLs found in powersh...

Classification

Based on following Playbook procedure and the analysis of available information, this alert is determined to be a successful True Positive.

Report:

Incident Summary

Analysis confirms a successful True Positive alert for Emotet malware on the host Maxim (172[.]16.17.83). The malicious document, MES 2020_12_31 S632974.doc, was allowed to execute on the endpoint. Network logs verify the execution was successful, showing subsequent outbound C2 traffic to three unique external IP addresses.

Incident Details

- **Alert Name:** SOC109 - Emotet Malware Detected
- **Target Host:** Maxim (172[.]16.17.83)
- **Flagged File:** MES 2020_12_31 S632974.doc
- **File Hash:** eee99e6d8ade9463dd206dfbad3485ea
- **Device Action:** Allowed

Analysis and Evidence

- **Execution:** The initial delivery method is unclear from the available logs, but the device action confirms the malicious document was allowed to execute. Sandbox analysis indicates the document utilises embedded macros and PowerShell to initiate its routines.
- **Network Activity:** Following execution, firewall logs show the host establishing outbound HTTP GET requests to three unique external IP addresses:
 - o 23[.]111.174.153 requesting `hxxp://decpak[.]com/cgi-bin/gU/`
 - o 152[.]170.79.100 requesting `hxxp://152[.]170.79.100/076ay2uof/umojx2x1vf1qjjk/hue4e670x/d3eobn8z0k0rp/syaxabx0loj/erz7hayf/`
 - o 190[.]247.139.101 requesting `hxxp://190[.]247.139.101/i3u3l3e3n96/sow63klj/pkv3runqw/dudwqjl4zg8l7hk6ah6/huulsuajy/nibyefksabf1mz63/`

Threat Intelligence

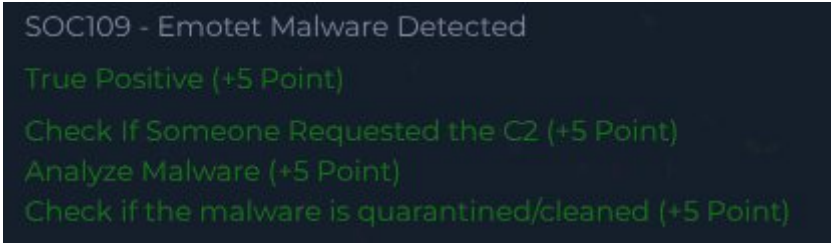
- **OSINT (Payload):** VirusTotal definitively flags the document as malicious (51/63 vendors) and categorises it as a Trojan downloader belonging to the Emotet family.
- **Sandbox Analysis:** Joe's Sandbox classifies the file as malicious with 100% confidence, highlighting multi-AV detection, malicious macro behaviour, and anomalous PowerShell execution.

Actions Taken

1. Verified the file hash via OSINT (VirusTotal), confirming it as an Emotet payload.
2. Reviewed sandbox telemetry (Joe's Sandbox), confirming the document employs malicious macros and PowerShell to reach out to external domains.
3. Correlated sandbox behaviours with internal network logs, identifying three successful outbound connections to malicious infrastructure.

4. Contained endpoint Maxim.
5. Declared True Positive and escalated the alert.

Let's Defend Result



```
SOC109 - Emotet Malware Detected
True Positive (+5 Point)
Check If Someone Requested the C2 (+5 Point)
Analyze Malware (+5 Point)
Check if the malware is quarantined/cleaned (+5 Point)
```